

SEAH SHAO XUAN

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Quantitative researcher and execution specialist with experience optimizing systematic equity trading strategies. Strong expertise in the complexities of APAC markets, particularly in execution impact, signal monetization, and trading optimization. Proficient in leveraging data, statistical modeling, and low-latency infrastructure to generate robust and stable returns.

SKILLS

Programming Languages:	Python, SQL, kdb q, C/C++, R, Shell Scripting
Data and Machine Learning:	PyTorch, TensorFlow, Hugging Face Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV
Big Data and Cloud:	Spark, Hadoop, HDFS, Hive, YARN, Airflow

EDUCATION

Carnegie Mellon University	Pittsburgh, PA, USA
Master of Science in Machine Learning, School of Computer Science (4.00/4.00)	Aug '23 to Aug '24
• Generative Models and Image Synthesis	• Distributed ML for Large Datasets
• Variational Inference and MC Sampling	• (Full stack) Deep Learning Systems
National University of Singapore	Singapore
Bach. of Eng. (Mech. Eng.) (4.79/5.00) and Bach. of SocSci. (Economics) (4.81/5.00)	Aug '18 to May '22
NUS Merit Scholarship, Minor in Computer Science and Minor in Statistics	
• Machine Learning and Deep Learning	• Econometrics, Time Series, Causal Inference
• Multi-agent Reinforcement Learning	• Microeconomics, Game Theory, Auctions

EXPERIENCE

Squarepoint Capital	Singapore
Desk Quantitative Analyst	Nov '24 to PRESENT
• Conducted execution research and alpha monetization for systematic equity mid-frequency strategies across APAC markets, focusing on improving signal realization and enhancing performance under diverse liquidity and microstructure conditions	
• Led evaluation and integration of alpha signals into production trading frameworks, optimizing for net monetization after transaction costs, market impact, and execution constraints, while actively identifying and mitigating alpha decay and underexposure within market segments	
• Designed and optimized portfolio and execution risk parameters to improve risk-adjusted returns, enhance robustness across volume and volatility segments, and reduce unintended factor and liquidity exposures under changing market conditions	
• Developed and implemented statistical market impact and cost models directly embedded into live execution logic, improving execution quality and reducing trading costs for high-throughput systematic strategies	
• Designed and implemented market-specific trading constraints and execution logic to balance regulatory compliance, liquidity access, and execution efficiency while preserving strategy intent across complex APAC markets	
• Improved production-grade trading systems for live strategies, and built scalable tooling systems for execution analysis and diagnostics, enabling rapid identification of performance degradation and faster deployment of strategy enhancements	
Sea Labs, Sea Limited	Singapore
Data Engineer	Jun '22 to Jul '23
• Optimized distributed data computation within the Spark-Hadoop ecosystem for petabytes of financial data, reducing cluster-memory resource consumption of tasks by 80%	
• Developed CI/CD testing and pipelines for operational continuity and task accuracy	
• Developed in-house tools for operational diagnostics and monitoring	
Temasek Labs, National University of Singapore	Singapore
Computer Vision Research Assistant	Jan '21 to May '22
• Developed real-time small object detection system by utilising ensembles of correlation filters	
• Developed real-time algorithm for 3D reconstruction of a spacetime curve using camera invariants	
• Led software operationalisation and deployment in lightweight embedded systems for field use	

PUBLICATIONS

[1] Seah Shao Xuan, Sutthiphong Srigrarom (2023). Multiple UAS Traffic Planning Based on Deep Q-Network with Hindsight Experience Replay and Economic Considerations. MDPI Aerospace.

[2] Seah Shao Xuan, Lau Yan Han, Sutthiphong Srigrarom (2022). Multiple Aerial Targets Re-identification by 2D and 3D Kinematics-based Matching. MDPI Imaging.